

AA2-ServidorDHCP

1-Primer canviem la ip del segon adaptador modificant-lo l'arxiu, podem accedir amb "sudo nano /etc/netplan/(ubicació de l'arxiu)"

I ho posarem així l'adaptador de xarxa intern:

enp0s9:

dhcp4: no

addresses: [192.168.22.1/24]

```
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:70:c8:5e brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.22/24 brd 192.168.56.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe70:c85e/64 scope link
        valid_lft forever preferred_lft forever
```

2-Instal·lem kea amb "sudo apt install kea-dhcp4-server kea-dhcp6-server kea-admin -y"

3-Desactivarem el DHCPv6 i DDNS amb aquestes 2 ordres:

sudo systemctl stop kea-dhcp6-server

sudo systemctl disable kea-dhcp6-server

```
usuari@server22:~$ sudo systemctl disable kea-dhcp6-server
Synchronizing state of kea-dhcp6-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable kea-dhcp6-server
usuari@server22:~$
```

4-Ara crearem un fitxer nou per a kea amb:

cd /etc/kea

sudo mv kea-dhcp4.conf old-kea-dhcp4.conf

sudo nano kea-dhcp4.conf

Ara serem a dins i ho hem de configurar, per configurar-ho posem aquest text modificat els camps necessaris.

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "enp0s9" ]
    },

    "valid-lifetime": 4000,
    "renew-timer": 1000,
    "rebind-timer": 2000,

    "lease-database": {
      "type": "memfile",
      "persist": true,
      "name": "/var/lib/kea/kea-leases4.csv"
    },

    "subnet4": [
      {
```

```

"subnet": "192.169.22.0/24",
"option-data": [
  {
    "name": "routers",
    "data": "192.169.22.254"
  },
  {
    "name": "domain-name-servers",
    "data": "8.8.8.8"
  }
],

"pools": [
  {
    "pool": "192.169.22.10-192.169.22.50"
  }
],

"reservations": [
  {
    "hw-address": "08:00:27:57:20:fc",
    "ip-address": "192.169.22.20"
  }
]
}
]
}
}
}

```

Ara apliquem la configuració reiniciant el servidor amb
 "sudo systemctl restart kea-shcp4-server"

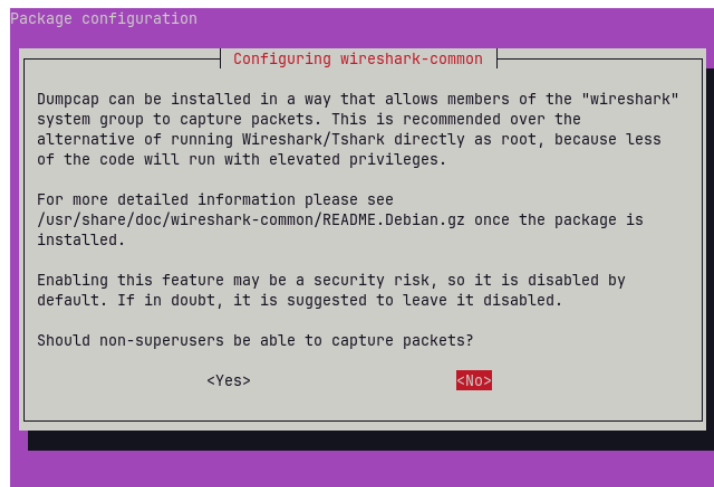
Ara comprovar que funciona amb "sudo systemctl status kea-dhcp-server"

```

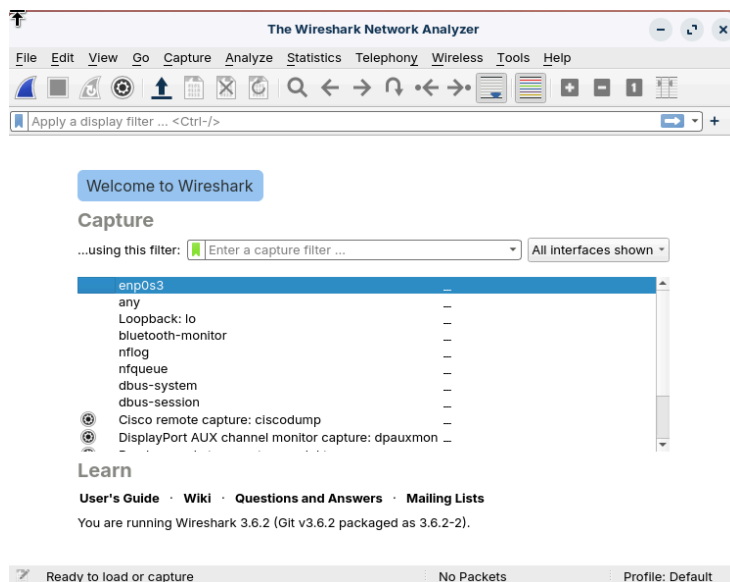
usuari@server22:~$ sudo systemctl status kea-dhcp4-server
• kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-02 18:12:09 UTC; 4s ago
     Docs: man:kea-dhcp4(8)
  Main PID: 1838 (kea-dhcp4)
    Tasks: 7 (limit: 4605)
   Memory: 3.5M (peak: 3.9M)
      CPU: 17ms
   CGroup: /system.slice/kea-dhcp4-server.service
           └─1838 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

```

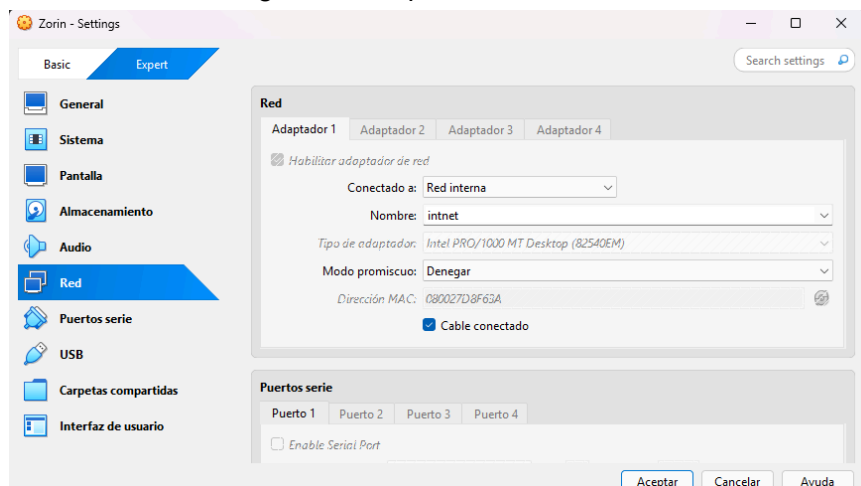
5-Instal·lem wireshark a la màquina del client amb "sudo apt install wireshark"



I ho obrim amb "sudo wireshark"



6-Iniciem la captura de trànsit donant-li a l'aleta Anem ara a la configuració i el posem en xarxa interna



Ara refresquem la ip desconectant i connectant la configuració de xarxa dins de zorin

Esperem molt poc i ja parem la captura de trànsit
I ara passem a analitzar els resultats

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000980000	10.0.2.15	146.190.225.48	TCP	74	42710 → 80 [SYN] Seq=9 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=4088971519 TS...
2	0.003366103	146.190.225.48	10.0.2.15	TCP	60	80 → 42710 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
3	0.003419939	10.0.2.15	146.190.225.48	TCP	54	42710 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
4	0.003602911	10.0.2.15	146.190.225.48	HTTP	137	GET / HTTP/1.1
5	0.003795865	146.190.225.48	10.0.2.15	TCP	60	80 → 42710 [ACK] Seq=1 Ack=84 Win=65535 Len=0
6	0.006964255	146.190.225.48	10.0.2.15	HTTP	303	HTTP/1.1 200 OK (text/plain)
7	0.006989414	10.0.2.15	146.190.225.48	TCP	54	42710 → 80 [ACK] Seq=84 Ack=250 Win=63991 Len=0
8	0.007089172	10.0.2.15	146.190.225.48	TCP	54	42710 → 80 [FIN, ACK] Seq=84 Ack=250 Win=63991 Len=0
9	0.007300565	146.190.225.48	10.0.2.15	TCP	60	80 → 42710 [ACK] Seq=250 Ack=85 Win=65535 Len=0
10	0.008369193	146.190.225.48	10.0.2.15	TCP	60	80 → 42710 [FIN, ACK] Seq=250 Ack=85 Win=65535 Len=0
11	0.008381394	10.0.2.15	146.190.225.48	TCP	54	42710 → 80 [ACK] Seq=85 Ack=251 Win=63991 Len=0
12	5.338429187	PCSSystemtec_7a:d9...	52:55:0a:00:02:02	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
13	5.338552521	52:55:0a:00:02:02	PCSSystemtec_7a:d9...	ARP	64	10.0.2.2 is at 52:55:0a:00:02:02
14	19.294567811	fe80::fa4e:843:b927...	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
15	19.442442062	fe80::fa4e:843:b927...	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
16	19.593168548	PCSSystemtec_7a:d9...	Broadcast	ARP	42	Who has 10.0.2.3? Tell 10.0.2.15
17	19.615777976	10.0.2.15	224.0.0.251	MDNS	189	Standard query response 0x0000 PTR, cache flush user-VirtualBox.local AAAA, c...
18	19.620875156	10.0.2.15	224.0.0.251	MDNS	189	Standard query response 0x0000 PTR, cache flush user-VirtualBox.local AAAA, c...

- 1–10 TCP / HTTP 10.0.2.15 ↔ 146.190.225.48 Unicast
- 12 ARP PCSSystemtec_7a:d9... → Broadcast Broadcast (MAC)
- 13 ARP 10.0.2.15 → PCSSystemtec_7a:d9... Unicast
- 14–15 ICMPv6 fe80::fa4e... → ff02::16 Multicast (Broadcast IP)
- 16 ARP Broadcast → PCSSystemtec_7a:d9... Broadcast (MAC)
- 17–18 MDNS 10.0.2.15 → 224.0.0.251 Multicast (Broadcast IP)